

# An Introduction to $p$ -Adic Numbers

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# Constructing $p$ -Adic Numbers

- Let  $p$  be a prime number.
- Any rational number can be expressed in base  $p$  as

$$\begin{aligned}x_M &= p^\gamma \sum_{j=0}^M x_j p^j \\ &= p^\gamma (x_0 + x_1 p + x_2 p^2 + \cdots + x_M p^M),\end{aligned}$$

- where the digits satisfy  $x_j \in \{0, 1, \dots, p-1\}$ , the leading digit  $x_0 \neq 0$ ,  $\gamma \in \mathbb{Z}$ , and  $M \in \mathbb{N}$ .

# The field of $p$ -Adic Numbers

- By allowing the number of digits to grow without bound, we obtain a  $p$ -adic number:

$$\lim_{M \rightarrow \infty} x_M = p^\gamma \sum_{j=0}^{\infty} x_j p^j$$

- Finally, we can construct the field of  $p$ -Adic Numbers as:

$$\mathbb{Q}_p := \left\{ p^\gamma \sum_{j=0}^{\infty} x_j p^j \mid x_j \in \{0, 1, \dots, p-1\}, \gamma \in \mathbb{Z} \right\}.$$

# The $p$ -Adic Norm

- Let's define the **norm**, a function, as:

$$|\cdot| : \mathbb{Q} \rightarrow \mathbb{R}_{\geq 0}$$

satisfying:

- $|x| = 0 \iff x = 0$ ,
  - $|xy| = |x| |y|$ ,
  - $|x + y| \leq |x| + |y|$  (triangle inequality).
- The  **$p$ -adic norm**, denoted  $|\cdot|_p$ , is defined as follows:

$$x \in \mathbb{Q}^\times, \quad x = p^\gamma \frac{a}{b}, \quad \gcd(a, p) = \gcd(b, p) = 1 \implies |x|_p = p^{-\gamma}$$

$$|0|_p = 0$$

# The $p$ -Adic Norm

- The norm  $|\cdot|_p$  is **non-Archimedean**:

$$|x + y|_p \leq \max\{|x|_p, |y|_p\}.$$

- A  $p$ -adic number can be obtained as the limit of a sequence of rational numbers when we measure distances using the  $p$ -adic norm,  $|\cdot|_p$ .
- Note that  $\mathbb{Q}_p$  is the completion of  $\mathbb{Q}$  with respect to the  $p$ -adic norm,  $|\cdot|_p$ .

# The Ring of $p$ -Adic Integers

**Definition:** The *ring of  $p$ -adic integers*, denoted  $\mathbb{Z}_p$ , is defined as

$$\mathbb{Z}_p = \{ x \in \mathbb{Q}_p \mid |x|_p \leq 1 \}.$$

- A  $p$ -adic integer is simply a  $p$ -adic number whose absolute value does not exceed 1. This means that its  $p$ -adic expansion contains no negative powers of  $p$ .

$$x = x_0 + x_1p + x_2p^2 + \cdots, \quad x_i \in \{0, 1, \dots, p-1\}.$$

- A **metric space** is a pair  $(M, d)$ , where  $M$  is a set and  $d$  is a function that assigns a distance between any two points in  $M$ . Formally,  $d : M \times M \rightarrow \mathbb{R}_{\geq 0}$  must satisfy the following properties for all  $a, b, c \in M$ :

①  $d(a, b) = 0$  if and only if  $a = b$ .

②  $d(a, b) = d(b, a)$  (symmetry).

③  $d(a, b) \leq d(a, c) + d(c, b)$  (triangle inequality).



- A metric space  $(M, d)$  is called an *ultrametric* space if  $d$  satisfies

$$d(a, b) \leq \max\{d(a, c), d(c, a)\} \quad \text{for any } a, b, c \in M.$$

- **Theorem:**  $\mathbb{Q}_p$  with the distance  $d(x, y) = |x - y|_p$  is a complete ultrametric space.

- For any  $a \in \mathbb{Q}_p$  and integer  $r$ , we define:

$$B_r(a) := \{x \in \mathbb{Q}_p : |x - a|_p \leq p^r\}.$$

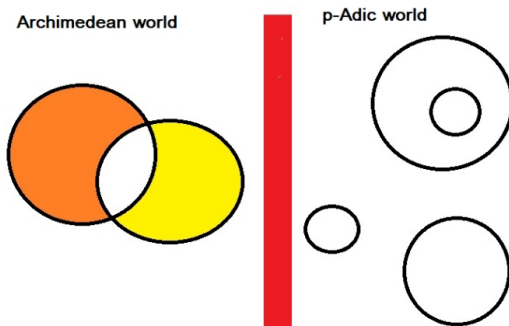
$$S_r(a) := \{x \in \mathbb{Q}_p : |x - a|_p = p^r\}.$$

To be the ball and sphere with center  $a$  and radius  $p^r$ .

# Topology of $\mathbb{Q}_p$

Key properties of  $p$ -adic balls:

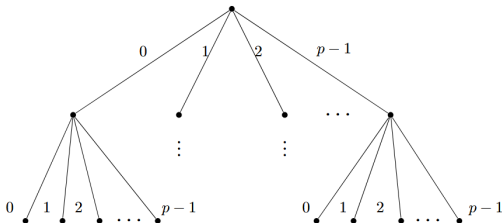
- 1 Any point inside a ball is also a center of that ball.
- 2 Every ball is simultaneously **open** and **closed**.
- 3 Two balls in  $\mathbb{Q}_p$  are disjoint or one contains the other.



# Applications

- Let's consider the following expansion

$$x = x_0 + x_1p + x_2p^2 + \cdots + x_{\ell-1}p^{\ell-1} \quad x_i \in \{0, 1, \dots, p-1\}.$$

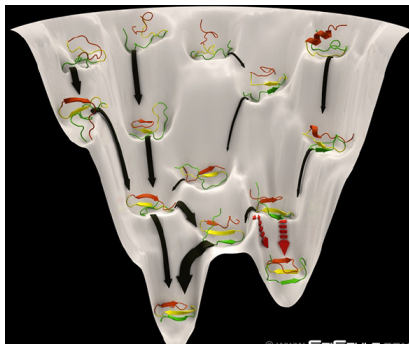


- The root corresponds to “no digits chosen yet”.
- The first level chooses  $x_0$  (there are  $p$  possible children).
- The second level chooses  $x_1$ , and so on.
- For a fixed depth  $\ell$ , the set of truncations

$$G_\ell := \{(x_0, x_1, \dots, x_{\ell-1}) : x_i \in \{0, 1, \dots, p-1\}\}$$

forms a **finite subtree**.

# The energy landscape



- Many complex systems (for example, proteins) can be described as moving on an **energy landscape**.
- Each point in the landscape represents a possible configuration of the system.
- Low regions correspond to **stable states** (local minima), while high regions correspond to **unstable states**.

# From landscapes to trees

- A complex landscape with many valleys can be approximated by a **hierarchical tree of basins (or pools)**.
- Each valley (local minimum) is a small basin; as we raise the energy level, nearby basins merge into larger ones.
- The resulting tree has an **ultrametric structure**, very similar to the finite rooted trees used for  $p$ -adic numbers and the graphs  $G_\ell$ .



Paisaje de energía

Grafo de desconectividad  
(espacio ultramétrico asociado)

# References



Jack A. Thorne.

*p-adic Analysis and p-adic Arithmetic.*

Tutorial notes, Harvard University, Summer 2010.

<https://www.dpmms.cam.ac.uk/~jat58/all.pdf>.



Alexa Pomerantz.

*An Introduction to the p-Adic Numbers.*

University of Chicago REU (2020).

[https:](https://math.uchicago.edu/~may/REU2020/REUPapers/Pomerantz.pdf)

[//math.uchicago.edu/~may/REU2020/REUPapers/Pomerantz.pdf](https://math.uchicago.edu/~may/REU2020/REUPapers/Pomerantz.pdf).



W. A. Zúñiga-Galindo.

*Ultrametricity: Theory and Applications.*

Graduate Seminar, UTRGV (2021).



Oscar Francisco Casas Sánchez, Leonardo Fabio Chacón Cortés, Jeanneth Galeano Peñaloza.

*Una introducción al análisis p-ádico.*

Manuscript, Universidad Nacional de Colombia.

Thank you for your time